

1 Metric TSP

In the metric traveling salesperson problem (TSP) we are given a connected graph $G = (V, E)$ with edge costs $c_e \geq 0$ for all $e \in E$, and the goal is to find the minimum cost tour that visits all vertices at least once and returns to the starting point.

To obtain such a tour, it is enough to select a minimum cost multi-set of edges from E which form a connected, Eulerian graph. Such graphs have an Eulerian tour: a walk on the edges so that every edge is used exactly once. Note that on such a walk, the starting and ending vertices must be the same since every vertex has even degree, so these walks are TSP tours.

This is equivalent to an alternate formulation of the problem where we are given a metric over the point set and the goal is to find the cheapest Hamiltonian cycle (hence the name metric TSP). The equivalence can be seen by shortcutting: if $v_1, v_2, \dots, v_1$ is the sequence of vertices visited by the Eulerian tour, one can remove all but the first appearance of every vertex in the sequence (except v_1 , for which we keep the first and final appearance). By the triangle inequality, this Hamiltonian cycle is no longer than the initial tour. To see the equivalence in the other direction, note we can use the shortest path metric.

1.1 Approximating Metric TSP

This is an NP-Hard problem, and in fact is APX-Hard: there is no approximation algorithm for metric TSP with ratio below about 1.01 unless $P=NP$ [KLS15]. There is a $3/2$ approximation from the 70s due to Christofides and, independently, Serdyukov, and the best known approximation is just slightly better than that, which we'll discuss later in the course.

A natural special case of this problem is when the input graph is unweighted, i.e., all edges have the same cost. This is called the graphic case. Then the goal becomes simpler: find the connected Eulerian multi-subgraph with as few edges as possible. Here, we will discuss the following theorem of Vishnoi [Vis12]:

Theorem 1.1. *Let G be a simple k -regular connected graph. Then there is a connected, spanning Eulerian multi-subgraph with at most $(1 + O(\frac{1}{\log k}))n$ edges.*

Vishnoi actually proved a slightly weaker guarantee, with a $\sqrt{\log k}$ instead. The streamlined analysis we discuss here is due to [CV15].

This result leads to an approximation algorithm with ratio tending to 1 as $k \rightarrow \infty$, since every TSP tour has at least n edges. In other words, graphic TSP on k -regular graphs becomes "easy" as k grows. We remark that follow-up work demonstrates that there is always a tour with at most $(1 + O(\frac{1}{k}))n$ edges [Chr+25b], and a $1 + O(\frac{\log k}{k})$ algorithmic approximation [Chr+25a].

1.2 An Algorithmic Idea

A classic algorithmic idea for graphic TSP is as follows.

1. Pick a random cycle cover: a collection of vertex disjoint cycles that cover the whole graph. This graph is Eulerian, but not connected.
2. To fix the connectivity, we add a spanning tree on the graph resulting from contracting the cycles. So that this does not destroy the Eulerian property, then double all the edges of the spanning tree.

The cycle cover always has n edges. If there are m cycles, the doubled spanning tree has $2m - 2 \leq 2m$ edges. So, if there are m cycles, the approximation ratio is at most

$$\frac{n + 2m}{n} = 1 + 2m/n$$

We will show that for a random cycle cover, the expected number of cycles shrinks as k grows, so that the expected approximation ratio tends to 1 as $k \rightarrow \infty$. In particular, we will show the expected number of cycles is at most $O(\frac{n}{\log k})$, leading to [Theorem 1.1](#).

Consider the directed graph $\tilde{G}$ arising from bidirecting every edge of G . A cycle cover in a directed graph is a collection of directed cycles, possibly of length 2. Our distribution over cycle covers will be the uniform distribution over cycle covers in $\tilde{G}$. After sampling the directed cycles, we simply drop the orientation of the edges. If a cycle has length 2, it leads to a doubled edge.

In the remainder, we will show the following two lemmas:

Lemma 1.2. *There is a randomized polynomial time algorithm which samples a uniformly random cycle cover from $\tilde{G}$, up to polynomially small total variation distance $\epsilon > 0$.*

We will show how this is a corollary of the randomized $1 + \epsilon$ approximation for the permanent of Jerrum, Sinclair, and Vigoda [[JSV04](#)] that runs in time polynomial in $1/\epsilon$ which we mentioned in a previous lecture, however we will not show the randomized approximation for the permanent itself.

Lemma 1.3. *Let X measure the number of cycles in a uniformly random cycle cover sampled from $\tilde{G}$. Then with probability at least $1 - e^{-n}$, X has at most $O(\frac{n}{\log k})$ cycles.*

These finish the proof. The idea is simple. We apply the first lemma to obtain a uniformly random cycle cover up to polynomially small total variation distance $\epsilon = \frac{1}{n}$. Combining the second lemma with this error, we obtain a cycle cover with at most $O(\frac{n}{\log k})$ cycles with probability at least $1 - 1/n - e^{-n} \geq 1 - 2/n$. Otherwise, we obtain a cover with at most $\frac{n}{2}$ cycles, the maximum possible. It follows that our expected number of cycles is at most:

$$O(\frac{n}{\log k}) + \frac{n}{2} \cdot \frac{2}{n} \leq O(\frac{n}{\log k})$$

as desired.

1.3 Sampling a Cycle Cover

Create a bipartite graph from G as follows. Put one copy of each vertex on the left side and one on the right. Then, for every edge $\{u, v\}$ add two edges to the bipartite graph, one from the left copy of u to the right copy of v and one from the left copy of v to the right copy of u .

If we select a perfect matching in this graph, every vertex in the original graph will have degree exactly two (one edge from its left copy and one from its right copy, possibly the same edge twice), giving us a cycle cover. Look at the $M \in \{0,1\}^{V \times V}$ matrix resulting from the above bipartite graph. Then, the permanent of this matrix counts the number of perfect matchings and therefore the number of cycle covers of $\tilde{G}$.

Let's prove the lemma using the randomized approximation for the permanent.

Proof. Consider the edges of the bipartite graph one by one in an arbitrary order $e_1, \dots, e_{|E|}$. For each edge $e_i = \{u, v\}$, compute the probability that e_i is in the perfect matching by constructing two new bipartite graphs with corresponding matrices: M^+ , in which e_i is selected, where we delete u, v and all adjacent edges, and M^- , in which e_i is not selected, where we simply delete e_i . Let the subscript H indicate the previous choices of the algorithm, corresponding to deleting nodes or edges. Then if M is the final matching and we condition on the decisions of $e_1, \dots, e_{i-1}$:

$$\mathbb{P}[e_i \in M \mid e_1, \dots, e_{i-1}] = \frac{\text{per}(M_H^+)}{\text{per}(M_H^-) + \text{per}(M_H^+)}.$$

Using the $1 + \epsilon$ approximation for the permanent, we estimate this probability up to $\epsilon = 1/n^3$ accuracy (with probability exponentially close to 1, which for simplicity we will treat as probability 1). We then flip a coin with this probability to determine if we select e_i or not. Once we select this edge, we move to the graph with matrix M^+ or M^- accordingly. Now, suppose we can compute the permanent exactly at each step. Then, the probability of some particular cycle cover is:

$$\mathbb{P}_{S \sim \mu}[e_1 \in S, e_2 \notin S, \dots, e_{|E|} \in S] = \mathbb{P}[e_1 \in S] \mathbb{P}[e_2 \notin S \mid e_1 \in S] \dots \mathbb{P}[e_{|E|} \in S \mid e_1 \in S, \dots]$$

as needed.

However, every step has some polynomial error, each cycle cover accrues some error over the steps of the algorithm, and there are exponentially many covers. So how do we achieve $1/n$ total variation distance? The trick is to couple the "perfect" version of this sampler, which outputs a uniformly random cycle cover, with our imperfect version. By sharing the same randomness for each coin toss, the two versions differ with probability at most $n^2 \cdot 1/n^3 = 1/n$ by the union bound, since there are at most n^2 total edges. $\square$

1.4 Properties of a Random Cycle Cover

If we scale M down so that there is $\frac{1}{k}$ on every non-zero entry, then the matrix is doubly stochastic. Therefore, its permanent is at least e^{-n} by the previous lecture. Scaling back, we multiply the permanent by k^n , implying that there are at least $k^n e^{-n}$ perfect matchings.

The key intuition for the algorithm is this:

Lemma 1.4. *The number of cycle covers in $\tilde{G}$ with at least m cycles is at most $2^n k^{n-m}$.*

Proof. We will demonstrate a way to generate every cycle cover. First, choose a set S of m vertices. Then, for every vertex not in S , pick an outgoing edge. There are at most 2^n choices for S and k^{n-m} choices for the remaining edges. If the collection of edges form a set of disjoint paths and cycles, close each path by adding an edge from the end to the beginning.

In such a way, every cycle cover containing at least m cycles can be generated: given such a cycle cover, delete one edge in m of the cycles. The vertices with no outgoing edge are S and will be recovered. This proves the claim. $\square$

But now the idea is that there are actually at least $k^n e^{-n}$ cycle covers. $k^n e^{-n}$ dominates $2^n k^{n-m}$ for m some function that grows slowly with k . Let's prove [Lemma 1.3](#):

Proof. Applying the above lemma with $m = \frac{3n}{\log k}$, there are at most

$$2^n k^{n - \frac{3n}{\log k}} = 2^n k^n e^{-3n} \leq k^n e^{-2n}$$

covers with at least $\frac{3n}{\log k}$ cycles. Each cover is chosen with probability at most the inverse of the number of perfect matchings in the bipartite graph, so at most $\frac{1}{k^n e^{-n}}$. Therefore the total probability mass of these covers is at most:

$$k^n e^{-2n} \frac{1}{k^n e^{-n}} \leq e^{-n}$$

It follows that with probability at least $1 - e^{-n}$, we choose a cover with fewer than $\frac{3n}{\log k} = O(\frac{n}{\log k})$ cycles, as desired. $\square$

References

- [Chr+25a] Micha Christoph, Nemanja Draganić, António Girão, Eoin Hurley, Lukas Michel, and Alp Müyesser. *Cycle-Factors of Regular Graphs via Entropy*. 2025. DOI: [10.48550/arXiv.2507.19417](#). arXiv: [2507.19417 \[math.CO\]](#) (cit. on p. 1).
- [Chr+25b] Micha Christoph, Nemanja Draganić, António Girão, Eoin Hurley, Lukas Michel, and Alp Müyesser. *New Bounds for Linear Arboricity and Related Problems*. 2025. DOI: [10.48550/arXiv.2507.20500](#). arXiv: [2507.20500 \[math.CO\]](#) (cit. on p. 1).
- [CV15] Ashish Chiplunkar and Sundar Vishwanathan. “Approximating the Regular Graphic TSP in Near Linear Time”. In: *35th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS)*. Vol. 45. Leibniz International Proceedings in Informatics. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2015, pp. 125–135. DOI: [10.4230/LIPIcs.FSTTCS.2015.125](#) (cit. on p. 1).
- [JSV04] Mark Jerrum, Alistair Sinclair, and Eric Vigoda. “A Polynomial-Time Approximation Algorithm for the Permanent of a Matrix with Nonnegative Entries”. In: *Journal of the ACM* 51.4 (2004), pp. 671–697. DOI: [10.1145/1008731.1008738](#) (cit. on p. 2).
- [KLS15] Marek Karpinski, Michael Lampis, and Richard Schmied. “New Inapproximability Bounds for TSP”. In: *Journal of Computer and System Sciences* 81.8 (2015), pp. 1665–1677. DOI: [10.1016/j.jcss.2015.06.003](#) (cit. on p. 1).
- [Vis12] Nisheeth K. Vishnoi. “A Permanent Approach to the Traveling Salesman Problem”. In: *Proceedings of the 53rd Annual IEEE Symposium on Foundations of Computer Science (FOCS)*. IEEE, 2012, pp. 76–80. DOI: [10.1109/FOCS.2012.81](#) (cit. on p. 1).